Supplementary Table 4: Summary of included studies based on current diagnostic criteria for autoimmune encephalitis, management and clinical outcomes

Study	Case no.	Age	Sex	Criteria 1[a]:			Criteria 2:				Criteria 3:	Diagnostic certainty	Neuronal Antibody	AEACN anatomical classification[b]	Treatment	Treatment Response
				Altered sensorium	Working memory deficits	Neuropsychiatric symptoms	New focal CNS findings	Seizures	MRI suggestive of encephalitis	CSF pleocytosis	Exclusion of alternate etiology					
Ayatollahi et al, 2020 (15)	1	18	F		✓		✓	✓	✓	✓	✓	Possible	Negative	Subcortical	MP	Yes
Ayuso et al, 2020 (16)	2	72	F	✓			✓				✓	Probable Bickerstaff's brainstem encephalitis	GD1a	Cerebellar	MP	Yes
Burr et al, 2020 (17)	3	2	F	✓			✓			✓	✓	Possible	NMDA-R	N.A.	MP, IVIg	Yes
Allahyari et al, 2021 (18)	4	18	F			✓		✓		✓	✓	Possible	NMDA-R	N.A.	IVIg	Yes
Guilmot et al 2020 (71)	5	62	M	✓	✓		✓		✓		✓	Probable Bickerstaff's brainstem encephalitis	GD1b	N.A.	IVIg	Yes
	6	80	M		✓	✓		✓		✓	✓	Possible	CASPR2	N.A.	MP, PLEX	Yes
	7	71	N	✓			✓				✓	Probable	GD1b	N.A.	N.A.	NA

			A								Bickerstaff's brainstem encephalitis				
Panariello et al (19)	8	23	M	✓		✓	✓			✓	✓	Possible	NMDA-R	N.A.	Dexamethasone, IVIg	Yes
Jeanneret et al, 2021[d] (20)	9	21	M				✓		✓		✓	Definite	GAD-65	Combined	MP, IVIg, PLEX followed by 2nd course of IVIg followed by PLEX	Yes
Khoo et al, 2020 (63)	10	65	F			✓	✓				✓	Possible	Negative	N.A.	MP	Yes
Zambreanu et al, 2020 (21)	11	66	F	✓	✓	✓		✓	✓		✓	Possible	Negative	Limbic	MP. IVIg, oral steroids	Yes
Pizzanelli et al , 2021 (22)	12	76	F	✓				✓	✓		✓	Possible	Negative	Limbic	MP then oral steroids	Yes
Monti et al et al, 2020 (23)	13	50	M			✓	✓	✓		✓	✓	Definite	NMDA-R	N.A.	MP, IVIg, PLEX	Yes
Mulder et al, 2021 (24)	14	40	M	✓			✓		✓[e]		✓	Definite	Novel Antibody	N.A.	PLEX then MP	Yes
Parnasa et al, 2021 (25)	15	56	M	✓		✓				✓	✓	Possible	Negative	N.A.	MP	Yes
Pilotto et		68	M	✓				✓	✓		✓	Possible	Negative	Limbic	N.A.	Yes

al, 2020 (68)	16														
	17	76	F	✓					✓	✓	✓	Possible	Negative	N.A.	N.A.	Yes
	18	74	F	✓				✓	✓	✓	✓	Possible	Negative	N.A.	N.A.	Exp
	19	60	M	✓				✓	✓	✓	✓	Possible	Negative	Limbic	N.A.	Exp
	20	60	F	✓						✓	✓	Possible	Negative	N.A.	N.A.	Yes
	21	61	M	✓			✓			✓	✓	Possible	Negative	N.A.	N.A.	Yes
	22	52	M	✓			✓			✓	✓	Possible	Negative	N.A.	N.A.	Yes
Sanchez et al, 2021 (5)	23	61	F	✓				✓	✓	✓	✓	Definite	NMDA-R	Limbic	MP, PLEX	Yes
	24	53	F	✓			✓		✓		✓	Possible	Negative	Combined	MP, PLEX	N
	25	62	M	✓				✓			✓	Possible	Negative	N.A.	No immunotherapy	Yes
	26	63	M	✓			✓		✓		✓	Possible	Negative	Brainstem	No immunotherapy	Yes
	27	46	F	✓							✓	Possible	Negative	N.A.	No immunotherapy	Yes

Sarigecili et al 2021 (26)	28	7	M	✓			✓	✓			✓	Definite	NMDA-R	N.A.	MP, IVIg, oral steroids	Yes
McHattie et al , 2021 (27)	29	53	F	✓			✓	✓	✓	✓	✓	Definite	NMDA-R	Combined	IVIg	Yes
Sánchez-Morales et al, 2021 (28)	30	14	M	✓			✓	✓			✓	Definite	NMDA-R	N.A.	MP IVIg	N
Hosseini et al, 2021 (29)	31	46	M	✓			✓	✓			✓	Possible	Negative	N.A.	No immunotherapy	Yes
Kilic et al, 2021[d] (30)	32	3	F				✓	✓			✓	Definite	CASPR-2	Encephalomyelitis	IVIg MP	Yes
Paolo Manganotti et al 2021 (44)	33	37	M	✓			✓	✓			✓	Possible	Negative	N.A.	IVIg	Yes
	34	71	M	✓			✓				✓	Possible	Negative	N.A.	IVIg	Yes
Yousuf et al, 2021 (31)	35	60	M	✓							✓	Possible	Negative	Combined	IVIg	Yes
Gaughan et al, 2021 (32)	36	16	F			✓					✓	Definite	GAD-65	Limbic	IVIg MP	Yes
Moideen et al, 2020 (33)	37	17	M			✓					✓	Definite	NMDA-R	N.A.	MP	Yes
Asan et al 2021 (34)	38	44	M				✓				✓	Definite	GFAP	N.A.	MP	Yes

Valadez-Calderon et al 2021 (35)	39	28	M	✓			✓		✓		✓	Definite	NMDA-R, GAD-65	Limbic	MP	Yes
Dono et al 2020 (36)	40	81	M	✓			✓		✓	✓	✓	Possible	Negative	Cotical/sub cortical	MP	N
Bravo G et al 2021 (37)	41	30	F	✓		✓		✓	✓	✓	✓	Definite	NMDA-R	Limbic	MP, IVIg	Yes
Bhagat,al 2021 (38)	42	54	M	✓				✓	✓		✓	Possible	Negative	Limbic	No immunoth erapy	Yes
Smyth et al, 2021 (39)	43	50	M	✓			✓				✓	Definite	LGI1	Non contrbutory	MP	Yes
Zuhorn et al 2020 (59)	44	54	M			✓	✓			✓	✓	Possible	Negative	Subcortical	N.A.	Yes
Oosthuizen K et al. 2021 (58)	45	52	M			✓	✓			✓	✓	Possible	Ampiphys in	Braintem	MP	Yes
Franke et al. 2021 (69)	46	69	M	✓			✓			✓	✓	Definite	NMDA-R	N.A.	N.A.	Na
	47	58	F	✓						✓	✓	Definite	Myelin	N.A.	N.A.	
	48	54	M	✓			✓	✓			✓	Definite	Myelin	N.A.	N.A.	
	49	58	F	✓			✓			✓	✓	Definite	Yo	N.A.	N.A.	Na
Delorme et		72	M			✓	✓		✓	✓	✓	Possible	Negative	N.A.	IVIg, MP	Yes

al. 2020 (67)	50														
	51	66	F			✓			✓		✓	Possible	Negative	N.A.	MP	Yes
	52	60	F			✓			✓		✓	Possible	Negative	N.A.	MP	Yes
	53	69	M			✓		✓	✓		✓	Possible	Negative	N.A.	MP	Yes
Vendervorst et al.2020 (40)	54	29	M	✓	✓				✓		✓	Possible	Negative	Limbic	No immunotherapy	Yes
Khawagi et al. 2020 (41)	55	7	F	✓		✓	✓	✓			✓	Possible	Negative	N.A.	No immunotherapy	Yes
El-Zein et al. 2020 (55)	56	40	M	✓	✓	✓					✓	Possible	Negative	N.A.	IVIg	Yes
Baretta et al. 2021 (56)	57	64	M	✓		✓	✓		✓		✓	Possible	Negative	Cortical/subcortical	MP IVIg	Yes
Ishaq et al 2021 (56)	58	63	M	✓		✓					✓	Possible	Negative	N.A.	MP, IVIg	Yes
Gunawardhane et al. 2021 (53)	59	47	F	✓				✓			✓	Possible	Negative	Combined	MP IVIg	Yes
Le guennec at al (42)	60	69	M	✓					✓		✓	Possible	Negative	Combined	IVIg	Yes
Cao et al. 2021 (6)	61	49	M	✓			✓		✓		✓	Possible	Negative	Combined	MP PLEX	Yes

	62	56	M	✓					✓		✓	Possible	Negative	Combined	MP PLEX	Yes
	63	61	M	✓			✓		✓		✓	Possible	Negative	Combined	MP	Yes
	64	37	M	✓					✓	✓	✓	Possible	Negative	Combined	MP	n
	65	77	F	✓					✓		✓	Possible	Negative	Cortical/subcortical	MP	n
Fiere-Alvarez et al. 2020 (60)	66	39	M	✓					✓	✓	✓	Possible	Negative	Combined	IVIg, Tocilizumab	Yes
Afshar et al 2020 (61)	67	39	F	✓				✓	✓		✓	Possible	Negative	Combined	IVIg MP	n
Kremer et al. 2020 (70)	68	71	F	✓		✓			✓		✓	Possible	Negative	Subcortical	No immunotherapy	na
	69	64	M	✓		✓			✓		✓	Possible	Negative	Brainstem	No immunotherapy	na
Benameur et al, 2020 (73)	70	34	M	✓			✓		✓		✓	Possible	Negative	Subcortical	N.A.	N.A.
	71	64	M	✓			✓		✓		✓	Possible	Negative	Limbic	No immunotherapy	Yes
Hara et al. 2021 (54)	72	65	M	✓		✓					✓	Possible	Negative	N.A.	MP, oral steroids, IVIg	Yes
Arzia-		48	F	✓		✓			✓		✓	Possible	Negative	Limbic	MP	Yes

varon et al. 2021(62)	73															
Urso et al. 2020 (43)	74	5	F	✓		✓				✓	✓	Possible	Negative	Cortical/subcortical	Dexamethasone	Yes

a-Criteria of possible autoimmune encephalitis as per Graus et al. 2016
b- Autoimmune Encephalitis Alliance Clinicians Network proposed anatomico-clinical classification for autoimmune encephalitis, Abboud et al.2021
c- early onset delirium with akathisia, choreiform involuntary movements of the upper limbs and gait ataxia
d-Clinical profile is consistent with GAD-65 positivity, hence included despite not fulfilling the criteria
e- PET was consistent with novel antibody
Note: Cases 2,5,7 partially fulfilled the criteria for probable Bickerstaff brainstem encephalitis by Graus et al. None of the patients had external ophthalmoplegia. As per AEACN proposed classification, these patients can represent brainstem form of autoimmune encephalitis
N.A. Not applicable, Exp: expired, MP: Methylprednisolone, IVIg: Intravenous immunoglobulin